

Oral exam — question bank

TIF285 / FYM285, Learning from data, Fall 2026

Student version

Your envelope holds one question card with three questions from this bank: one on Project 1 and one on Project 2 (one technical, one evaluative), and an extension question about a situation the projects did not cover. Q1 and Q2 are worth 0–6 points each; Q3 is worth 0–6 for variant (a) and 0–8 for variants (b) and (c). The example on the next page shows how to read a card. See the oral exam instructions on Canvas.

The two projects

Project 1

Bayesian parameter estimation for chiral EFT: the three-nucleon couplings c_D , c_E from few-nucleon observables, with an eigenvector-continuation emulator.

Project 2

Deep neural network emulator for the chaotic three-body problem (Breen et al.).

Example question set

This is an example, not an actual exam set. It shows a card exactly as it is printed; the questions are taken from the bank below.

TIF285 / FYM285 ORAL EXAMINATION — QUESTION SET — EXAMPLE
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Read all three questions. You have at least 30 minutes to prepare. Your preferred project is discussed first. Write the set number on your handover card.

PROJECT 1

The prior on each LEC was $\mathcal{N}(0, 5^2)$. What would have changed in your Case 1 and Case 2 posteriors had you used a prior width of 50 instead, or of 0.5?

PROJECT 2

Your training and validation curves coincide. A colleague says “so the network generalises well”. Do you agree?

EXTENSION — prepare only the variant the assistant has circled

If you did both extra tasks, your variant is **(c)**.

- (a) No extra task.** Both projects build an emulator of an expensive simulator — eigenvector continuation in Project 1, a neural network in Project 2. What do they have in common, and how do their error characteristics differ?
- (b) Project 1 extra task (inferring Q).** How did Q enter your likelihood, and what did its posterior look like? Your Project 1 analysis treated the emulator as exact. Suppose the emulator error were comparable to σ_{EFT} . What would inferring Q then do — what would Q absorb, and how would you detect that it had?
- (c) Project 2 extra task (Bayesian neural network).** How did you split the predictive variance into two terms, and which was larger? The eigenvector-continuation emulator in Project 1 carries no error bar either. Which of your two terms is the right analogue for an EC emulator, and how would you actually obtain it?

How to read this card:

- Your preferred project is discussed first (Q1), the other project second (Q2).
- Of the extension question, prepare only the variant the assistant circles: (a) if you did no extra task, the one for your extra task if you did one, and the one printed on the card — here (c) — if you did both.

Part A — Technical questions

Reconstruct the method; say what the terms do; get the details right.

Project 1: Theme 1 — Prior, likelihood, posterior

[T1.1] Each single-observable posterior in Task 2 came out as a narrow, curved band in the (c_D, c_E) plane. The posterior is the product of a likelihood and a prior: which of the two decides its shape across the band, and which along it? What does that tell you about a MAP point read off a single-observable posterior?

[T1.2] The prior on each LEC was $\mathcal{N}(0, 5^2)$. What would have changed in your Case 1 and Case 2 posteriors had you used a prior width of 50 instead, or of 0.5?

[T1.3] A posterior conditioned on a single observable comes out as a narrow curved band in the (c_D, c_E) plane rather than a peak. Why?

[T1.4] Task 2 evaluated the posterior on a grid; task 3 used MCMC. What exactly goes wrong with the grid, and when does it become hopeless?

[T1.5] What did your prior predictive in Task 1 show, and what does it mean that the prior looks weakly informative in parameter space but not in observable space?

[T1.6] The prior predictive put a large fraction of its mass on unphysical binding energies. Does that mean the prior was wrong, and did it matter for the rest of the analysis?

Project 1: Theme 2 — The error model

[T2.1] The total variance was $\sigma^2 = \sigma_{\text{exp+method}}^2 + \sigma_{\text{EFT}}^2$. Justify adding variances. What assumption is buried there?

[T2.2] Compare Fig. 6a and 6b. What happened when the truncation term was added, and why?

[T2.3] Compare the σ_{EFT} values you computed in Task 2 with $\sigma_{\text{exp+method}}$. What does that comparison say about where the uncertainty in this analysis comes from, and what would be worth improving first?

Project 1: Theme 3 — MCMC and convergence

[T3.1] What is the sampler actually doing? Describe one proposed step.

[T3.2] How did you decide burn-in? What does a trace plot show, and what can it not show?

[T3.3] What is the autocorrelation time and why does it govern how many samples you really have?

[T3.4] Your corner plot shows a strong c_D - c_E correlation. What does it mean, and what does it do to sampling?

Project 1: Theme 4 — Predictions and information content

[T4.1] Your posterior predictive in Task 4 was built from the posterior samples. What does its width represent, and what would a prediction for a new, independent measurement of $E(^4\text{He})$ need in addition?

[T4.2] Two parameters, and naively two data would do. Case 1 used $\{E(^4\text{He}), R_p(^4\text{He})\}$, Case 2 all four observables. What changed between the two cases, and what did the extra data buy?

[T4.3] Is any single observable more informative about c_D than the others, and how can you tell?

[T4.4] The emulator takes milliseconds per evaluation; the full many-body solver takes about a minute. How long would your Task 3 have taken with the full solver, and what would you have done differently?

Project 2: Theme 5 — Data, pipeline and architecture

[T5.1] Your network maps the time t and the initial position of particle 2 directly to the positions at time t , rather than stepping forward from the positions at the previous time. What does that design gain, and what does it lose, compared with a network that predicts the next time step?

[T5.2] Particle 3 is not an output of the network. Suppose you had kept all three particles' positions as outputs anyway: what would the network then have to learn, and how could you check whether it had learned it?

[T5.3] The collision trajectories were removed before training. What would a network trained on them have learned, and which of your later results would that have affected?

[T5.4] The split into training and validation data was made between whole trajectories. Would a row-by-row split have made a measurable difference for your network? Why, or why not, and when would it matter?

Project 2: Theme 6 — Training and its diagnostics

[T6.1] Describe one training step: what happens between reading a batch and updating the weights.

[T6.2] Your training and validation curves lie on top of each other. What does that tell you, and what does it *not* tell you?

[T6.3] Your validation error came out slightly *below* the training error. Bug, leakage, or neither?

[T6.4] What is over- and underfitting, and how would each show in your learning curves?

Project 2: Theme 7 — Emulation: cost, accuracy and validity

[T7.1] Your network predicts a trajectory in about a millisecond; Brutus takes minutes to hours. Under what circumstances is the network actually the cheaper option, and what does the speed-up number leave out?

[T7.2] Where do both networks fail, and is that a property of the training or of the problem?

[T7.3] The data stop at $t = 3.9$. Can the network be asked for $t = 8$? What happens, and how would you know whether to believe it?

Project 2: Theme 8 — Chaos and what an emulator can deliver

[T8.1] Describe what you did in Task 5 and what the hundred points did as time increased. What does that tell you about the system, and about how well any emulator can ever do here?

[T8.2] Once the hundred points have spread out, is the emulator still useful, and for what? Your own, less accurate network gave roughly the same spread as the pre-trained one — is that expected?

Part B — Evaluative questions

Take a position on the analysis and defend it. No single right answer; what is assessed is whether the judgement is specific and checkable.

Project 1

[E1] Which part of your Project 1 analysis do you trust least, and what would it take to convince you it is sound?

[E2] The analysis assumes the errors on the four observables are independent. Is that a good assumption here, and what would you expect if it were relaxed?

[E3] Your posterior came out narrower than the paper's in some directions. Is that a success or a warning sign, and how would you tell?

[E4] How much of your result is driven by the data and how much by the prior? How did you find out, or how would you?

[E5] The emulator replaces a 60-second calculation with a millisecond one. Where in parameter space would you expect it to be least accurate, and why? How would you check that its approximation does not affect your posterior?

[E6] The posterior predictive for one observable sits several σ from experiment. What are the candidate explanations, in what order, and how would you separate them?

[E7] If you had to drop one of the four observables, how would you decide which one, and what would you lose?

[E8] You checked your predictions against the same data used for calibration. Why is that a weak test, and what would be a strong one?

Project 2

[E9] Which part of your Project 2 analysis do you trust least, and what would it take to convince you it is sound?

[E10] Your training and validation curves coincide. A colleague says "so the network generalises well". Do you agree?

[E11] The error of your network varies by orders of magnitude between trajectories. What single number, or figure, would you use to summarise its accuracy honestly, and why not the mean MAE over the validation set?

[E12] The network gives a point prediction. Without building a Bayesian network, how could you attach an error bar to its predictions using what you already have, and what would that error bar miss?

[E13] Suppose your trained network disagreed badly with the pre-trained one on a trajectory that the pre-trained one handles well. List the explanations in the order you would check them.

Part C — Extension questions (Q3)

Reason about a situation the project did not cover: “what would change if...”, “why would this fail when...”. Each set has three variants, labelled on the card: (a) for students who did no extra task, worth 0–6 points; (b) and (c) for students who did the Project 1 or the Project 2 extra task, worth 0–8 points. These open with a short question on that extra task. A student who did both gets the variant printed on the card.

X1 — Emulators and their errors

(a) No extra task. Both projects build an emulator of an expensive simulator — eigenvector continuation in Project 1, a neural network in Project 2. What do they have in common, and how do their error characteristics differ?

(b) Project 1 extra task (inferring Q). How did Q enter your likelihood, and what did its posterior look like? Your Project 1 analysis treated the emulator as exact. Suppose the emulator error were comparable to σ_{EFT} . What would inferring Q then do — what would Q absorb, and how would you detect that it had?

(c) Project 2 extra task (Bayesian neural network). How did you split the predictive variance into two terms, and which was larger? The eigenvector-continuation emulator in Project 1 carries no error bar either. Which of your two terms is the right analogue for an EC emulator, and how would you actually obtain it?

X2 — Model discrepancy

(a) No extra task. Project 1 quantifies its model discrepancy explicitly. What is the corresponding quantity in Project 2, and why is it so much harder to state?

(b) Project 1 extra task (inferring Q). What did the Beta(3,5) prior encode, and how much did the data move it? In Project 2 there is no formula for the emulator’s error. Could you use the same device — a parametrised error model with a prior, inferred with the rest? What would the parameter be, and what would constrain it?

(c) Project 2 extra task (Bayesian neural network). Why does the notebook call the second uncertainty term “discrepancy” rather than “aleatoric”? Compare it with σ_{EFT} in Project 1. What is being learned in each case, what plays the role of the prior, and which of the two would you trust more, and why?

X3 — Data and information

(a) No extra task. Suppose the adopted experimental error on one observable were ten times smaller. What happens to the posterior on (c_D, c_E) , and what happens to the posterior predictive?

(b) Project 1 extra task (inferring Q). With Q inferred the sampler explores three dimensions — what changed for the (c_D, c_E) posterior compared with fixed Q ? Suppose the experimental error on one observable were ten times smaller. Does the posterior on Q change? Why, or why not?

(c) Project 2 extra task (Bayesian neural network). What happened to the epistemic

and the discrepancy terms as you went from 20 to 500 trajectories? In Project 1 more data narrows the posterior. Here the epistemic band did not shrink. Is that a contradiction?

X4 — Checks and validation

(a) No extra task. How would you tell whether your network has learned the physics or memorised the training set — and what is the analogous question in Project 1?

(b) Project 1 extra task (inferring Q). How did you check that the three-parameter chain had converged? The posterior on Q is itself a check of the error model. What would it have meant if the posterior on Q had peaked near 1, or near 0?

(c) Project 2 extra task (Bayesian neural network). How did you check the calibration of the Bayesian network, and what did you find? Could you run the same calibration check on the posterior predictive of Project 1? What would it need, and why is it hard there?

X5 — When results disagree

(a) No extra task. Both projects reproduce a published paper. Suppose your result disagreed with the paper's. How would you proceed?

(b) Project 1 extra task (inferring Q). What did you conclude about Q from its posterior? Suppose the posterior on Q had peaked sharply at 0.6, far from the 0.33 you had fixed earlier. List the explanations in the order you would check them, and how you would separate them.

(c) Project 2 extra task (Bayesian neural network). What did the two uncertainty terms do beyond the end of the training data? A colleague uses your Bayesian network inside an inference at $t = 6$ and reports a tight credible interval. What went wrong, and which of your two terms should have warned them?

X6 — What is assumed before the data arrive

(a) No extra task. In Project 1 you could write down a prior. What plays that role in Project 2, and how is it expressed?

(b) Project 1 extra task (inferring Q). Why a Beta(3,5) prior on Q rather than uniform on $(0, 1)$? The $\mathcal{N}(0, 5^2)$ prior on the LECs was checked in observable space with a prior predictive. What would a prior predictive check for Q look like — what does Beta(3,5) imply about the size of the error bars — and would it have changed anything?

(c) Project 2 extra task (Bayesian neural network). What prior did you put on the weights, and why $\mathcal{N}(0, 2/n_{\text{in}})$ per layer? Project 1's prior was checked by pushing draws through the model. What does your weight prior imply about the *functions* the untrained network represents, and how would you check that?

X7 — Two kinds of spread

(a) No extra task. Task 5 of Project 2 shows a hundred trajectories from nearly identical initial conditions spreading apart. Is there an uncertainty in Project 1 that would remain even if the LECs were known exactly? What is the analogy?

(b) Project 1 extra task (inferring Q). With Q inferred, the posterior predictive widened — how much of its width was parameter uncertainty and how much error model? In Project 2 the spread of Task 5 exists for a perfect emulator. Which part of your Project 1 predictive is the analogue, and what would it mean to report the other part alone?

(c) Project 2 extra task (Bayesian neural network). Which dominated in your runs — the chaotic spread of Task 5 or the epistemic band of the Bayesian network — and at what time? What would it mean for a scientific conclusion if you reported one and not the other, and what is the Project 1 analogue of each?

X8 — A new observable

(a) No extra task. You are given a new observable with no estimate of its theoretical uncertainty. Can it be included in the analysis? What would you do?

(b) Project 1 extra task (inferring Q). What determines σ_{EFT} for an observable in the formula you used? A new observable arrives with no error estimate at all. You already infer Q . Can you infer this observable's error the same way — and what is different about one parameter for four observables versus one parameter for one?

(c) Project 2 extra task (Bayesian neural network). Your Bayesian network has four outputs that predict a variance for each position, the discrepancy term. How do they learn the size of the network's error, when no error is ever observed directly? Could the same idea supply the missing uncertainty for a new observable in Project 1? What would you need, and what would you be assuming?